Lid-Driven Cavity Flow: A Finite-Volume Implementation

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1 Problem Description

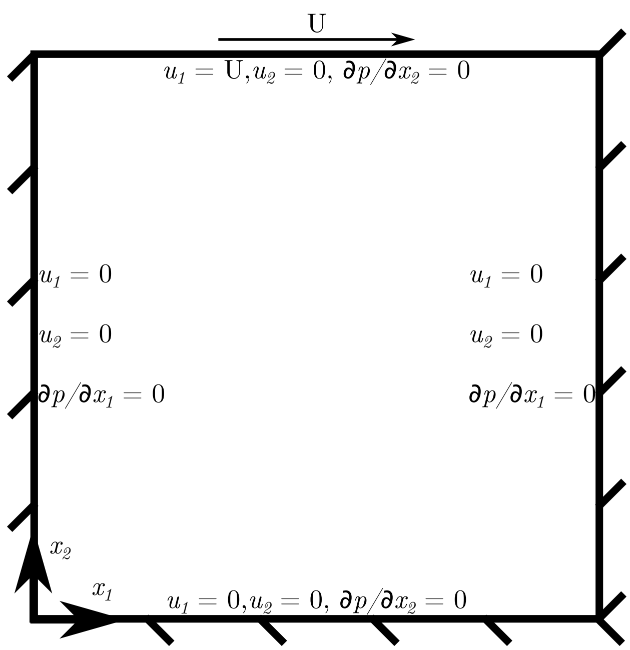

Figure 1: Lid-driven cavity flow problem illustration.

2 Governing Equations

2.1 Differential Equations

The form of the incompressible Navier-Stokes equations, and their subsequent non-dimensionalization are taken from Ferziger and Perić [1]. Dimensional quantities are shown with a "*" superscript, and non-dimensional quantities are shown without modification. Conservation of mass, with dimensional units is shown in Eq. 1. Similarly, the dimensional form of the momentum equation is given in Eq. 2.

$$\frac{\partial u_i^*}{\partial x_i^*} = 0 \tag{1}$$

$$\frac{\partial u_i^*}{\partial t^*} + \frac{\partial (u_j^* u_i^*)}{\partial x_j^*} = \nu^* \frac{\partial^2 u_i^*}{\partial x_j^2} - \frac{1}{\rho^*}\frac{\partial p^*}{\partial x_i^*} \tag{2}$$

The dimensional quantities of Eq. 1, 2, are nondimensionalized according to Eqs. 3 - 6.

$$t = \frac{t^*}{t_0^*} \tag{3}$$

$$x_i = \frac{x_i^*}{L_0^*} \tag{4}$$

$$u_i^* = \frac{u_i^*}{v_0^*} \tag{5}$$

$$p = \frac{p^*}{\rho^* v_0^{*2}} \tag{6}$$

Accordingly, the nondimensional form of the continuity and momentum equation are shown in Eq. 7 and 8, respectively. In these equations, L_0^* is the reference length dimension, t_0^* is the reference time dimension, v_0^* is the refence velocity dimension, and ρ^* is the density of the fluid.

$$\frac{\partial u_i}{\partial x_i} = 0 \tag{7}$$

$$\boxed{\mathrm{St}\frac{\partial u_i}{\partial t} + \frac{\partial (u_j u_i)}{\partial x_j} = \frac{1}{\mathrm{Re}}\frac{\partial^2 u_i}{\partial x_j} - \frac{\partial p}{\partial x_i}} \tag{8}$$

As a result of the nondimensionalization, the dimensionless numbers, Re (Reynolds number) and St (Strouhal number) appear in Eq. 8. While Eqs. 7 and 8 mathematically express mass and momentum conservation, respectively, with regards to implementing these numerically, the continuity equation (Eq. 7) does not provide any additional information that is not contained in the momentum equation (Eq. 8). The continuity equation must be further manipulated into a form which can be implemented to enforce mass continuity. Computing the divergence of the momentum equation (Eq. 8) yields Eq. 9. Simplifying Eq. 9 produces equation Eq. 10. Applying the contuinuity equation to Eq. 10 yeilds, Eq. 11, the pressure Poisson equation. In solving the pressure Poisson equation, mass conservation is enforced. Note that there is not a fluid equation of state for closure in which *absolute* pressure appears as a term. In incompressible flow, only *gradients* of pressure appear in the momentum equation, thus by solving Eq. 11, a pressure field which satisfies continuity is determined.

$$\frac{\partial}{\partial x_i}\left(\mathrm{St}\frac{\partial u_i}{\partial t}\right)+\frac{\partial}{\partial x_i}\left(\frac{\partial(u_j u_i)}{\partial x_j}\right)=\frac{\partial}{\partial x_i}\left(\frac{1}{\mathrm{Re}}\frac{\partial^2 u_i}{\partial x_j}\right)-\frac{\partial}{\partial x_i}\left(\frac{\partial p}{\partial x_i}\right) \tag{9}$$

$$\mathrm{St}\frac{\partial}{\partial t}\left(\frac{\partial u_i}{\partial x_i}\right)+\frac{\partial}{\partial x_i}\left(\frac{\partial(u_j u_i)}{\partial x_j}\right)=\frac{1}{\mathrm{Re}}\frac{\partial}{\partial x_j}\left(\frac{\partial}{\partial x_j}\left(\frac{\partial u_i}{\partial x_i}\right)\right)-\frac{\partial}{\partial x_i}\left(\frac{\partial p}{\partial x_i}\right) \tag{10}$$

$$\boxed{\frac{\partial}{\partial x_i}\left(\frac{\partial(u_j u_i)}{\partial x_j}\right)=-\frac{\partial}{\partial x_i}\left(\frac{\partial p}{\partial x_i}\right)} \tag{11}$$

2.2 Integral Equations

The integral forms of the non-dimensional momentum equation (Eq. 8) is required to apply a finite volume method numerical solution. The integral form of the non-dimensional momentum equation is obtained by integrating the differential form of the equation over a volume domain, Ω, which yields Eq. 12. Gauss's Divergence Theorem is applied to Eq. 12 to transform the volume integrals over a volume domain Ω into surface integrals over a surface boundary σ. Furthermore, the constant non-dimensional terms are moved outside of the integrals, producing Eq. 13. This integral form of the nondimensional pressure equation, Eq. 13 is used for solving control volume problem, and thus is intuitive for applying to the uniform two-dimensional cells of this problem. The non-dimensional pressure Poisson equation, Eq. 11, is left in differential form.

$$\int_\Omega \mathrm{St}\frac{\partial u_i}{\partial t}d\Omega+\int_\Omega \frac{\partial(u_j u_i)}{\partial x_j}d\Omega=\int_\Omega \frac{1}{\mathrm{Re}}\frac{\partial^2 u_i}{\partial x_j^2}d\Omega-\int_\Omega \frac{\partial p}{\partial x_i}d\Omega \tag{12}$$

$$\boxed{\mathrm{St}\int_\Omega \frac{\partial u_i}{\partial t}d\Omega+\int_\sigma u_j u_i n_j d\sigma=\frac{1}{\mathrm{Re}}\int_\sigma \frac{\partial u_i}{\partial x_j}n_j d\sigma-\int_\sigma p n_j d\sigma} \tag{13}$$

3 Discretization

3.1 Domain and Grid

The domain of the problem depicted in Fig. 1 is described in dimensional and non-dimensional quantities in Eqs. 14, 15, respectively. A staggered grid is implemented, such that pressure nodes are placed in the center of the cells ($x_1 = x_{1,M}$, $x_2 = x_{2,M}$), u_1 nodes are placed at the mid-height of the west and east faces of the cells ($x_1 = x_1$, $x_2 = x_{2,M}$), and u_2 nodes are placed at the mid-width of the south and north faces of the cells ($x_1 = x_{1,M}$, $x_2 = x_2$), where the subscript "M" indicates "mid-point". The grid is uniform, as described in

Table 1: Coordinate locations for different node types.

Node type	x_1	x_2
u_1	$m\Delta x$	$n\frac{\Delta x}{2}$
u_2	$m\frac{\Delta x}{2}$	$n\Delta x$
p	$m\frac{\Delta x}{2}$	$n\frac{\Delta x}{2}$

Eq. 16. An index "m" is used for the x_1 coordinate, and an index "n" is used for the x_2 coordinate. The coordinates of the the u_1, u_2, and pressure (p) nodes are tabulated in Table 3.1, for clarity, and the grid is illustrated in Fig. 2. There are ghost-nodes outside of the boundary, in both the x_1 and x_2 directions to enforce boundary conditions.

$$x_1^* \in [0, L_0^*] x_2^* \in [0, L_0^*] \tag{14}$$

$$x_1 \in [0, 1] x_2 \in [0, 1] \tag{15}$$

$$\Delta x \equiv \Delta x_1 = \Delta x_2 \tag{16}$$

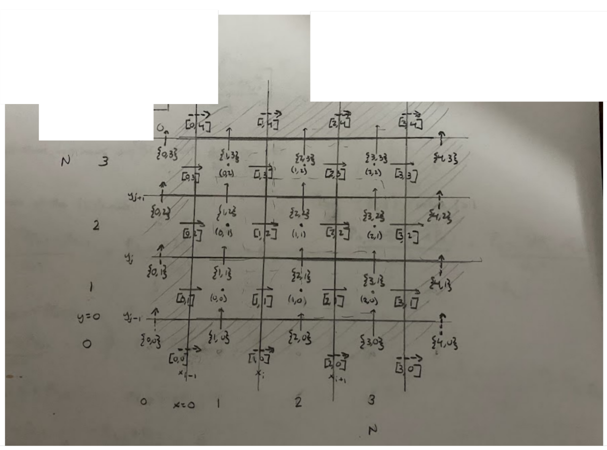

Figure 2: Illustration of staggered velocity-pressure grid.

3.2 Semi-Discrete Momentum Equation: Predictor and Corrector Step

The momentum equations, given in Eqs. 8, 13, cannot be solved in their given form as the pressure field that appears in these equations does not necessarily satisfy continuity. Accordingly, the momentum equation is split into a predictor step equation, Eq. 17, and a corrector step equation, Eq. 18, by discretizing the equations temporally. Note that "k" is the time-step index. An intermediate velocity field, u_i^*, is solved for in Eq. 17 which does not satisfy mass continuity. To obtain an equation for a pressure field which does enforce mass continuity, the divergence of Eq. 18 is computed and the divergence of u_i^{k+1} is set to 0, yielding Eq. 19, a Poisson equation for pressure. The corrector equation, Eq. 18 is then solved using the pressure solved for in Eq. 19, and a velocity field at time: $k+1$ is obtained which does satisfy mass continuity.

$$\mathrm{St}\frac{u_i^* - u_i^k}{\Delta t} = -\frac{\partial(u_j u_i)}{\partial x_j} + \frac{1}{\mathrm{Re}}\frac{\partial^2 u_i}{\partial x_j} \tag{17}$$

$$\mathrm{St}\frac{u_i^{k+1} - u_i^*}{\Delta t} = -\frac{\partial p}{\partial x_i} \tag{18}$$

$$\boxed{\mathrm{St}\frac{1}{\Delta t}\frac{\partial u_i^*}{\partial x_i} = \frac{\partial}{\partial x_i}\left(\frac{\partial p}{\partial x_i}\right)} \tag{19}$$

Gauss's theorem is applied to re-cast the differential forms of the predictor step momentum equation, Eq. 17, and the corrector step momentum equation, 18 into integral equations for a finite volume approach, shown in Eqs. 20 and 21, respectively. In these equations, n, is the unit normal vector. The pressure Poisson equation is kept in differential form. Observe that on the right hand sides of Eqs. 20, 21 a temporal index, k, has been assigned which indicates that and *explicit* temporal discretization scheme will be implemented, meaning that the velocity field solution from the k time step will be used to solve for the velocity field at time $k+1$. The pressure field is not assigned a temporal index as it is merely a scalr field variable that is used to enforce mass continuity.

$$\boxed{\mathrm{St}\int_\Omega \frac{u_i^* - u_i^k}{\Delta t}d\Omega = -\int_\sigma u_j^k u_i^k n_j d\sigma + \frac{1}{\mathrm{Re}}\int_\sigma \frac{\partial u_i^k}{\partial x_j} n_j d\sigma} \tag{20}$$

$$\boxed{\mathrm{St}\int_\Omega \frac{u_i^{k+1} - u_i^*}{\Delta t}d\Omega = -\int_\sigma p n_i d\sigma} \tag{21}$$

3.3 Finite Volume Method and Spatial Discretization

The integral forms of the predictor momentum equation and corrector momentum equation are discretized in a finite volume method implementation in which the equations are organized into unsteady terms, convective flux, diffusive flux, and pressure flux terms, then numerically approximated as follows.

3.3.1 u_1 Momentum Equation

The predictor and corrector u_1 momentum equations are obtained by substituting $i = 1$ in the predictor and corrector momentum equations, Eqs. 20, 21. The unsteady term in the u_1 predictor momentum equation is evaluated by assuming that the value of u_1 at the center of the two-dimensional u_1 cell is the average value of u_1 evaluated over the volume of the cell, as shown in Eq. 22, which is 2nd-order accurate [1].

$$\mathrm{St}\int_{\Omega}\frac{u_1^* - u_1^k}{\Delta t}d\Omega \approx \mathrm{St}\left[\frac{u_1^* - u_1^k}{\Delta t}\right]\Delta x_1 \Delta x_2 \equiv A_{u_1}[u_1^* - u_1^k] \tag{22}$$

The first term on the right hand side of the u_1 predictor equation represents the convective flux of velocity at the boundaries of the 2D cell, and is is evaluated on the σ_e (East), σ_w (West), σ_n (North), and σ_s (South) cell interfaces (walls), as shown in Eq. 23. These surface integrals are approximated by assuming that the velocities at the centers of the cell walls are the average values of the velocities evaluated over the entire surface, which is 2nd-order accurate [1]. Convective flux terms, such as $F_{c,e}^k$ are defined, and the convective flux integrals are evaluated ash shown in Eqs. 24 - 27. In the two-dimensional u_1 cells, the velocities are not defined at the centers of the cell faces and thus are obtained by linear interpolation, which has 2nd-order spatial accuracy [1], as shown in Eqs. 28 - 33.

$$\begin{aligned}\int_{\sigma} u_j^k u_i^k n_j d\sigma = \int_{\sigma} u_1^k u_1^k n_1 d\sigma + \int_{\sigma} u_2^k u_1^k n_2 d\sigma = \\ \int_{\sigma_e} u_1^k u_1^k d\sigma - \int_{\sigma_w} u_1^k u_1^k d\sigma + \int_{\sigma_n} u_2^k u_1^k d\sigma - \int_{\sigma_s} u_2^k u_1^k d\sigma\end{aligned} \tag{23}$$

$$\int_{\sigma} u_1^k u_1^k n_1 d\sigma \equiv F_{c,e}^k \approx u_{1,e}^k u_{1,e}^k \Delta x_2 \tag{24}$$

$$-\int_{\sigma_w} u_1^k u_1^k d\sigma \equiv F_{c,w}^k \approx -u_{1,w}^k u_{1,w}^k \Delta x_2 \tag{25}$$

$$\int_{\sigma_n} u_2^k u_1^k d\sigma \equiv F_{c,n}^k \approx u_{2,n}^k u_{1,n}^k \Delta x_1 \tag{26}$$

$$-\int_{\sigma_s} u_2^k u_1^k d\sigma \equiv F_{c,s}^k \approx -u_{2,s}^k u_{1,s}^k \Delta x_1 \tag{27}$$

$$u_{1,e}^k \approx \frac{u_{1,(m,n)}^k + u_{1,(m+1,n)}^k}{2} \tag{28}$$

$$u_{1,w}^k \approx \frac{u_{1,(m-1,n)}^k + u_{1,(m,n)}^k}{2} \tag{29}$$

$$u_{1,n}^k \approx \frac{u_{1,(m,n)}^k + u_{1,(m,n+1)}^k}{2} \tag{30}$$

$$u_{1,s}^k \approx \frac{u_{1,(m,n-1)}^k + u_{1,(m,n)}^k}{2} \tag{31}$$

$$u_{2,n}^k \approx \frac{u_{2,(m,n)}^k + u_{2,(m+1,n)}^k}{2} \tag{32}$$

$$u_{2,s}^k \approx \frac{u_{2,(m,n-1)}^k + u_{2,(m+1,n-1)}^k}{2} \tag{33}$$

The second term on the right hand side of the u_1 predictor equation represents the diffusive flux of velocity at the boundaries of the two-dimensional cell, and is evaluated on the σ_e (East), σ_w (West), σ_n (North), and σ_s (South) cell interfaces (walls), as shown in Eq. 34. As with the convective flux terms, the values of the velocity derivatives at the centers of the cell walls are assumed to be the average values of the velocity derivatives evaluated over the entire surface, which is 2nd-order accurate [1]. Diffusive flux terms, such as $F_{d,e}^k$ are defined and these surface integrals are evaluated as shown in Eqs. 35 - 38. The velocity derivatives at the cell face centers are approximated using 2nd-order accurate central differencing schemes [1], shown in Eqs. 39-42.

$$\begin{aligned}\frac{1}{\mathrm{Re}}\int_\sigma \frac{\partial u_1^k}{\partial x_j} n_j d\sigma = \frac{1}{\mathrm{Re}}\int_\sigma \frac{\partial u_1^k}{\partial x_1} n_1 d\sigma + \frac{1}{\mathrm{Re}}\int_\sigma \frac{\partial u_1^k}{\partial x_2} n_2 d\sigma = \\ \frac{1}{\mathrm{Re}}\int_{\sigma_e} \frac{\partial u_1^k}{\partial x_1} d\sigma - \frac{1}{\mathrm{Re}}\int_{\sigma_w} \frac{\partial u_1^k}{\partial x_1} d\sigma + \frac{1}{\mathrm{Re}}\int_{\sigma_n} \frac{\partial u_1^k}{\partial x_2} d\sigma - \frac{1}{\mathrm{Re}}\int_{\sigma_s} \frac{\partial u_1^k}{\partial x_2} d\sigma\end{aligned} \tag{34}$$

$$\frac{1}{\mathrm{Re}}\int_{\sigma_e} \frac{\partial u_1^k}{\partial x_1} d\sigma \equiv F_{d,e}^k \approx \frac{1}{\mathrm{Re}} \frac{\partial u_1}{\partial x_1}\bigg|_e^k \Delta x_2 \tag{35}$$

$$-\frac{1}{\mathrm{Re}}\int_{\sigma_w} \frac{\partial u_1^k}{\partial x_1} d\sigma \equiv F_{d,w}^k \approx -\frac{1}{\mathrm{Re}} \frac{\partial u_1}{\partial x_1}\bigg|_w^k \Delta x_2 \tag{36}$$

$$\frac{1}{\mathrm{Re}}\int_{\sigma_n} \frac{\partial u_1^k}{\partial x_2} d\sigma \equiv F_{d,n}^k \approx \frac{1}{\mathrm{Re}} \frac{\partial u_1}{\partial x_2}\bigg|_n^k \Delta x_1 \tag{37}$$

$$-\frac{1}{\mathrm{Re}}\int_{\sigma_s} \frac{\partial u_1^k}{\partial x_2} d\sigma \equiv F_{d,s}^k \approx -\frac{1}{\mathrm{Re}} \frac{\partial u_1}{\partial x_2}\bigg|_s^k \Delta x_1 \tag{38}$$

$$\frac{\partial u_1}{\partial x_1}\bigg|_e^k \approx \frac{u_{1,(m+1,n)}^k - u_{1,(m,n)}^k}{\Delta x_1} \tag{39}$$

$$\frac{\partial u_1}{\partial x_1}\bigg|_w^k \approx \frac{u_{1,(m,n)}^k - u_{1,(m-1,n)}^k}{\Delta x_1} \tag{40}$$

$$\left.\frac{\partial u_1}{\partial x_2}\right|_n^k \approx \frac{u^k_{1,(m,n+1)} - u^k_{1,(m,n)}}{\Delta x_2} \tag{41}$$

$$\left.\frac{\partial u_1}{\partial x_2}\right|_s^k \approx \frac{u^k_{1,(m,n)} - u^k_{1,(m,n-1)}}{\Delta x_2} \tag{42}$$

The u_1 predictor equation can now be re-written as shown in Eq. 43.

$$\boxed{A_u[u^*_{1,(m,n)} - u^k_{1,(m,n)}] = -[F^k_{c,e} + F^k_{c,w} + F^k_{c,n} + F^k_{c,s}] + [F^k_{d,e} + F^k_{d,w} + F^k_{d,n} + F^k_{d,s}]} \tag{43}$$

The unsteady term in the u_1 corrector equation is approximated identical to the unsteady term in the u_1 predictor equations, shown in Eq. 22. The only term on the right hand side of the u_1 corrector equation is the pressure flux term. Because a staggered grid is used in this problem, the pressure nodes exist at the center of the East and West cell wall faces. The pressure at the center of the cell walls is assumed to be the average pressure evaluated over the wntire surface, which is 2nd-order accurate [1]. Accordingly, the pressure flux is approximated and discretized as shown in Eqs. 44 - 46.

$$\int_\sigma p n_1 d\sigma = \int_{\sigma_e} p d\sigma - \int_{\sigma_w} p d\sigma \tag{44}$$

$$\int_{\sigma_e} p d\sigma \equiv F^k_{p,e} \approx p_e \Delta x_2 = p_{(m+1,n)} \Delta x_2 \tag{45}$$

$$-\int_{\sigma_w} p d\sigma \equiv F^k_{p,w} \approx -p_w \Delta x_2 = -p_{(m,n)} \Delta x_2 \tag{46}$$

The u_1 corrector momentum equation can now be re-written as shown in Eq. 47.

$$\boxed{A_u[u^{k+1}_{1,(m,n)} - u^*_{1,(m,n)}] = -[F^k_{p,n} + F^k_{p,s}]} \tag{47}$$

3.3.2 u_2 Momentum Equation

The predictor and corrector u_2 momentum equations, which are obtained by substituting $i = 2$ in the predictor and corrector momentum equations, Eqs. 20, 21, are approximated and discretized in a nearly identical way to those of u_1. The procedure for discritizing and approximating the terms in these equations is described in the same level of detail to provide clarity.

The unsteady term in the u_2 predictor momentum equation is evaluated by assuming that the value of u_2 at the center of the two-dimensional u_2 cell is the average value of u_2 evaluated over the volume of the cell, as shown in Eq. 48, which is 2nd-order accurate [1].

$$\mathrm{St}\int_{\Omega}\frac{u_2^*-u_2^k}{\Delta t}d\Omega \approx \mathrm{St}\left[\frac{u_{2,(m,n)}^*-u_{2,(m,n)}^k}{\Delta t}\right]\Delta x_1\Delta x_2 \equiv A_u[u_{2,(m,n)}^*-u_{2,(m,n)}^k] \tag{48}$$

The first term on the right hand side of the u_2 predictor equation represents the convective flux of velocity at the boundaries of the 2D cell, and is evaluated on the σ_e (East), σ_w (West), σ_n (North), and σ_s (South) surface boundaries (walls) of the cell, as shown in Eq. 49. These surface integrals are approximated by assuming that the velocities at the centers of the cell walls are the average values of the velocities evaluated over the entire surface, which is 2nd-order accurate [1]. Convective flux terms, such as $F_{c,e}^k$, are defined, and the convective flux surface integrals are evaluated as shown in Eqs. 50 - 53. In the two-dimensional u_2 cells, the velocites are not defined at the centers of the cell faces and thus are obtained by linear interpolation, which has 2nd-order spatial accuracy [1], shown in Eqs. 54 - 59.

$$\begin{aligned}\int_{\sigma}u_j^ku_2^kn_jd\sigma = \int_{\sigma}u_1^ku_2^kn_1d\sigma + \int_{\sigma}u_2^ku_2^kn_2d\sigma = \\ \int_{\sigma_e}u_1^ku_2^kd\sigma - \int_{\sigma_w}u_1^ku_2^kd\sigma + \int_{\sigma_n}u_2^ku_2^kd\sigma - \int_{\sigma_s}u_2^ku_2^kd\sigma\end{aligned} \tag{49}$$

$$\int_{\sigma_e}u_1^ku_2^kn_1d\sigma \equiv F_{c,e}^k \approx u_{1,e}^ku_{2,e}^k\Delta x_2 \tag{50}$$

$$-\int_{\sigma_w}u_1^ku_2^kd\sigma \equiv F_{c,w}^k \approx -u_{1,w}^ku_{2,w}^k\Delta x_2 \tag{51}$$

$$\int_{\sigma_n}u_2^ku_2^kd\sigma \equiv F_{c,n}^k \approx u_{2,n}^ku_{2,n}^k\Delta x_1 \tag{52}$$

$$-\int_{\sigma_s}u_2^ku_2^kd\sigma \equiv F_{c,s}^k \approx -u_{2,s}^ku_{2,s}^k\Delta x_1 \tag{53}$$

$$u_{2,e}^k \approx \frac{u_{2,(m,n)}^k+u_{2,(m+1,n)}^k}{2} \tag{54}$$

$$u_{2,w}^k \approx \frac{u_{2,(m-1,n)}^k+u_{2,(m,n)}^k}{2} \tag{55}$$

$$u_{1,e}^k \approx \frac{u_{1,(m,n)}^k+u_{1,(m,n+1)}^k}{2} \tag{56}$$

$$u_{1,w}^k \approx \frac{u_{1,(m-1,n-1)}^k+u_{1,(m-1,n)}^k}{2} \tag{57}$$

$$u_{2,n}^k \approx \frac{u_{2,(m,n)}^k + u_{2,(m,n+1)}^k}{2} \tag{58}$$

$$u_{2,s}^k \approx \frac{u_{2,(m,n-1)}^k + u_{2,(m,n)}^k}{2} \tag{59}$$

The second term on the right hand side of the u_2 predictor equation represents the diffusive flux of velocity at the boundaries of the 2D cell, and is evaluated on the σ_e (East), σ_w (West), σ_n (North), and σ_s (South) cell interfaces (walls), as shown in Eq. 60. As with the convective flux terms, the values of the velocity derivatives at the centers of the cell walls are assumed to be the average values of the velocity derivatives evaluated over the entire surface, which is 2nd-order accurate [1]. Diffusive flux terms, such as $F_{d,e}^k$ are defined, and these surface integrals are evaluated as shown in Eqs. 61 - 64. The velocity derivatives at the cell face centers are approximated using 2nd-order accurate central differencing schemes [1], shown in Eqs. 65 - 68.

$$\begin{aligned} \frac{1}{\mathrm{Re}} \int_\sigma \frac{\partial u_2^k}{\partial x_j} n_j d\sigma = \frac{1}{\mathrm{Re}} \int_\sigma \frac{\partial u_2^k}{\partial x_1} n_1 d\sigma + \frac{1}{\mathrm{Re}} \int_\sigma \frac{\partial u_2^k}{\partial x_2} n_2 d\sigma = \\ \frac{1}{\mathrm{Re}} \int_{\sigma_e} \frac{\partial u_2^k}{\partial x_1} d\sigma - \frac{1}{\mathrm{Re}} \int_{\sigma_w} \frac{\partial u_2^k}{\partial x_1} d\sigma + \frac{1}{\mathrm{Re}} \int_{\sigma_n} \frac{\partial u_2^k}{\partial x_2} d\sigma - \frac{1}{\mathrm{Re}} \int_{\sigma_s} \frac{\partial u_2^k}{\partial x_2} d\sigma \end{aligned} \tag{60}$$

$$\frac{1}{\mathrm{Re}} \int_{\sigma_e} \frac{\partial u_2^k}{\partial x_1} d\sigma \equiv F_{d,e}^k \approx \frac{1}{\mathrm{Re}} \frac{\partial u_2}{\partial x_1}\bigg|_e^k \Delta x_2 \tag{61}$$

$$-\frac{1}{\mathrm{Re}} \int_{\sigma_w} \frac{\partial u_2^k}{\partial x_1} d\sigma \equiv F_{d,w}^k \approx -\frac{1}{\mathrm{Re}} \frac{\partial u_2}{\partial x_1}\bigg|_w^k \Delta x_2 \tag{62}$$

$$\frac{1}{\mathrm{Re}} \int_{\sigma_n} \frac{\partial u_2^k}{\partial x_2} d\sigma \equiv F_{d,n}^k \approx \frac{1}{\mathrm{Re}} \frac{\partial u_2}{\partial x_2}\bigg|_n^k \Delta x_1 \tag{63}$$

$$-\frac{1}{\mathrm{Re}} \int_{\sigma_s} \frac{\partial u_2^k}{\partial x_2} d\sigma \equiv F_{d,s}^k \approx -\frac{1}{\mathrm{Re}} \frac{\partial u_2}{\partial x_2}\bigg|_s^k \Delta x_1 \tag{64}$$

$$\frac{\partial u_2}{\partial x_1}\bigg|_e^k \approx \frac{u_{2,(m+1,n)}^k - u_{2,(m,n)}^k}{\Delta x_1} \tag{65}$$

$$\frac{\partial u_2}{\partial x_1}\bigg|_w^k \approx \frac{u_{2,(m,n)}^k - u_{2,(m-1,n)}^k}{\Delta x_1} \tag{66}$$

$$\frac{\partial u_2}{\partial x_2}\bigg|_n^k \approx \frac{u_{2,(m,n+1)}^k - u_{2,(m,n)}^k}{\Delta x_2} \tag{67}$$

$$\frac{\partial u_2}{\partial x_2}\bigg|_s^k \approx \frac{u_{2,(m,n)}^k - u_{2,(m,n-1)}^k}{\Delta x_2} \tag{68}$$

The u_2 predictor equation can now be re-written as shown in Eq. 69.

$$\boxed{A_u[u^*_{2,(m,n)} - u^k_{2,(m,n)}] = -[F^k_{c,e} + F^k_{c,w} + F^k_{c,n} + F^k_{c,s}] + [F^k_{d,e} + F^k_{d,w} + F^k_{d,n} + F^k_{d,s}]} \tag{69}$$

The unsteady term in the u_2 corrector equation is approximated identical to the unsteady term in the u_2 predictor equation, shown in Eq. 48. The only term on the right hand side of the u_2 corrector equation is the pressure flux term. Because a staggered grid is used in this problem, the pressure nodes exist at the center of the North and South cell wall faces. The pressure at the center of the cell walls is assumed to be the average pressure evaluated over the entire surface, which is 2nd-order accurate [1]. Accordingly, the pressure flux is approximated and discretized as shown in Eqs. 70 - 72.

$$\int_\sigma p n_2 d\sigma = \int_{\sigma_n} p d\sigma - \int_{\sigma_s} p d\sigma \tag{70}$$

$$\int_{\sigma_n} p d\sigma \equiv F^k_{p,n} \approx p_n \Delta x_1 = p_{(m,n+1)} \Delta x_1 \tag{71}$$

$$-\int_{\sigma_s} p d\sigma \equiv F^k_{p,s} \approx -p_s \Delta x_1 = -p_{(m,n)} \Delta x_1 \tag{72}$$

The u_2 corrector momentum equation can now be re-written as shown in Eq. 73.

$$\boxed{A_u[u^{k+1}_{2,(m,n)} - u^*_{2,(m,n)}] = -[F^k_{p,n} + F^k_{p,s}]} \tag{73}$$

3.4 Boundary Conditions

The no-slip velocity boundary conditions, $u_1 = 0$ (or U at the top wall) and $u_2 = 0$, at the domain boundaries ($x_1 = 0, L$ and $x_2 = 0, L$) are enforced through "ghost nodes" placed just beyond the domain boundaries. The locations of these ghost nodes are tabulated in Table 3.4 for clarity. A negative velocity is applied at the ghost nodes such that the velocity at the domain boundaries is exactly 0 (or U at the top wall). This is demonstrated in Eqs. 74 - 77. Note that the equation for the velocity applied to the ghost nodes above the top boundary, Eq. 75, accounts for the non-zero velocity of the lid, U. The normal component of velocity is simply set to 0 at the boundaries. These boundary conditions are applied at the beginning of each time step before the predictor, pressure Poisson, and corrector equation procedures are applied.

$$u_{1,(m=0,n)} = -u_{1,(m=1,n)} \tag{74}$$

$$u_{1,(m=N+1,n)} = 2U - u_{1,(m=N,n)} \tag{75}$$

Table 2: Coordinate locations of ghost nodes.

Wall	x_1	x_2	n	m
Left	$-\frac{\Delta x}{2}$	$\in [-\frac{\Delta x}{2}, L+\frac{\Delta x}{2}]$	0	$\in [0, N+1]$
Right	$L+\frac{\Delta x}{2}$	$\in [-\frac{\Delta x}{2}, L+\frac{\Delta x}{2}]$	$N+1$	$\in [0, N+1]$
Bottom	$\in [-\frac{\Delta x}{2}, L+\frac{\Delta x}{2}]$	$-\frac{\Delta x}{2}$	$\in [0, N+1]$	0
Top	$\in [-\frac{\Delta x}{2}, L+\frac{\Delta x}{2}]$	$L+\frac{\Delta x}{2}$	$\in [0, N+1]$	$N+1$

$$u_{2,(m,n=0)} = -u_{2,(m,n=1)} \tag{76}$$

$$u_{2,(m,n=N+1)} = -u_{2,(m,n=N)} \tag{77}$$

3.5 Pressure Poisson Equation

The pressure equation given in Eq. 19 is re-arranged to match the form of Eq. 78, as shown in Eq. 79, following the convention of Ref. [2]. Expanding Eq. 79 and evaluating at a given pair of (m, n) indices yields Eq. 80. The right-hand side of Eq. 80 is set to the term $R_{m,n}$ for convenience; Eq. 81. The predictor-step velocity terms in Eq. 80 are solved with 2nd-order accurate central differencing schemes as described in Eq. 82, 83. The second derivatives of pressure on the left-hand side of the equation in Eq. 80 are also approximated to second-order spatial accuracy with central differencing schemes shown in Eqs. 84, 85. Following the convention of [2], we define the NxN matrix, L_{x_1} as the matrix representation of the Laplacian operator in the x_1 direction expressed in Eq. 84 as shown in Eq. 86. Because the problem domain is square ($N_{x_1} = N_{x_2} = N, \Delta x_1 = \Delta x_2 = \Delta x$), the matrix representation of the Laplacian operator in the x_2 direction, L_{x_2}, is identical to L_{x_1}, as shown in Eq. 87. Matrices I_{x_1} and I_{x_2}, which are identity matrices of size $(N_{x_1} = N)x(N_{x_1} = N)$ and $(N_{x_2} = N)x(N_{x_2} = N)$, are defined and shown in Eqs. 88, 89. This Poisson equation problem re-cast in matrix representation is shown in Eq. 90. In Eq. **??**, the "$\bigotimes$" symbol denoted the Kronecker product, and "vec" denotes vectorized forms of a matrix in which the vector is formed by stacking the columns of the matrix.

$$-\frac{\partial^2 \phi}{\partial x^2} = f \tag{78}$$

$$\boxed{-\frac{\partial}{\partial x_i}\left(\frac{\partial p}{\partial x_i}\right) = -\mathrm{St}\frac{1}{\Delta t}\frac{\partial u_i^*}{\partial x_i}} \tag{79}$$

$$-\left[\frac{\partial^2 p_{(m,n)}}{\partial x_{1,m}^2} + \frac{\partial^2 p_{(m,n)}}{\partial x_{2,n}^2}\right] = -\mathrm{St}\frac{1}{\Delta t}\left[\frac{\partial u_{1,(m,n)}^*}{\partial x_{1,m}^*} + \frac{\partial u_{2,(m,n)}^*}{\partial x_{2,n}{}^*}\right] \tag{80}$$

$$R_{m,n} = -\mathrm{St}\frac{1}{\Delta t}\left[\frac{\partial u^*_{1,(m,n)}}{\partial x^*_{1,m}} + \frac{\partial u^*_{2,(m,n)}}{\partial x_{2,n}{}^*}\right] \tag{81}$$

$$\frac{\partial u^*_{1,(m,n)}}{\partial x_{1,m}} \approx \frac{u_{1,(m+1,n)} - u_{1,(m,n)}}{\Delta x_1} \tag{82}$$

$$\frac{\partial u^*_{2,(m,n)}}{\partial x_{2,n}} \approx \frac{u_{2,(m,n+1)} - u_{1,(m,n)}}{\Delta x_2} \tag{83}$$

$$\frac{\partial^2 p_{(m,n)}}{\partial x^2_{1,m}} \approx \frac{p_{(m-1,n)} - 2p_{(m,n)} + p_{(m+1,n)}}{\Delta x_1^2} \tag{84}$$

$$\frac{\partial^2 p_{(m,n)}}{\partial x^2_{2,n}} \approx \frac{p_{(m,n-1)} - 2p_{(m,n)} + p_{(m,n+1)}}{\Delta x_2^2} \tag{85}$$

$$L_{x_1} = \begin{bmatrix} 1 & -1 & \ddots & \ddots & \ddots & \ddots & \ddots \\ -1 & 2 & -1 & \ddots & \ddots & \ddots & \ddots \\ \ddots & -1 & 2 & -1 & \ddots & \ddots & \ddots \\ \ddots & \ddots & \ddots & \ddots & \ddots & \ddots & \ddots \\ \ddots & \ddots & \ddots & -1 & 2 & -2 & \ddots \\ \ddots & \ddots & \ddots & \ddots & -1 & 2 & -1 \\ \ddots & \ddots & \ddots & \ddots & \ddots & -1 & 1 \end{bmatrix} \tag{86}$$

$$L_{x_2} = \begin{bmatrix} 1 & -1 & \ddots & \ddots & \ddots & \ddots & \ddots \\ -1 & 2 & -1 & \ddots & \ddots & \ddots & \ddots \\ \ddots & -1 & 2 & -1 & \ddots & \ddots & \ddots \\ \ddots & \ddots & \ddots & \ddots & \ddots & \ddots & \ddots \\ \ddots & \ddots & \ddots & -1 & 2 & -2 & \ddots \\ \ddots & \ddots & \ddots & \ddots & -1 & 2 & -1 \\ \ddots & \ddots & \ddots & \ddots & \ddots & -1 & 1 \end{bmatrix} \tag{87}$$

$$I_{x_1} = \begin{bmatrix} 1 & \ddots & \ddots \\ \ddots & 1 & \ddots \\ \ddots & \ddots & 1 \end{bmatrix} \tag{88}$$

$$I_{x_2} = \begin{bmatrix} 1 & \ddots & \ddots \\ \ddots & 1 & \ddots \\ \ddots & \ddots & 1 \end{bmatrix} \tag{89}$$

$$\left(\frac{1}{\Delta x_1^2} L_{x_1} \bigotimes I_{x_2} + I_{x_1} \bigotimes \frac{1}{\Delta x_2^2} L_{x_2} \right) \mathrm{vec}(p_{m,n}) = \mathrm{vec}(R_{m,n}) \tag{90}$$

The Poisson problem cast into matrix form, Eq. 90, is a linear system of the form $Ax = b$ and can be solved as such.

3.6 Temporal Discretization

4 Solution Procedure

5 Results

References

1. Ferziger, J. H. & Perić, M. *Computational Methods for Fluid Dynamics* 3rd ed. (Springer, Heidelberg, 2002).
2. Zhang, X. *MA615 Numerical Methods for PDEs Spring 2020 Lecture Notes* 2020.